HACKSPHERE 3.0 * OFFICIAL PROJECT SUBMISSION

Smart Energy Microgrid

& Demand Forecasting Engine

Autonomous 24-Hour Energy Dispatch, BESS Battery Storage Simulation, and Fail-Safe ML Telemetry deployed on Zoho Catalyst AppSail.

13.35 kW

MODEL ACCURACY (RMSE)

Random Forest + QuickML Ensemble

500 kWh

BESS BATTERY SIMULATION

80% Base SoC * 10-100% Safety Bounds

Zoho Catalyst

FULL-STACK DEPLOYMENT

AppSail Python 3.9 + Interactive Dashboard

Platform: Zoho Catalyst | Repository: github.com/Koushik554/Bits-Hackthon

Verified: test_eval.py SUCCESS

The Microgrid Challenge & Problem Context

THE OPERATIONAL CRISIS

Grid Overload & Renewable Intermittency

- **Peak Demand Spikes:**
  Sudden surges (>160 kW) destabilize local distribution transformers, leading to brownouts and steep utility penalty tariffs.
- **Renewable Mismatch:**
  Solar irradiance peaks midday when baseline demand is low, while evening hours (18:00-22:00) create critical grid stress.
- **Blind Load Switching:**
  Traditional microgrids switch reactively AFTER grid overload occurs, risking micro-grid tripping and battery degradation.
- **Third-Party API Outages:**
  Commercial cloud ML prediction endpoints can suffer network timeouts or quota exhaustion, crippling autonomous grid logic.

OUR HACKSPHERE 3.0 SOLUTION

Predictive Smart Grid Controller

- **24-Hour Predictive Horizon:**
  Forecasts hourly kW consumption before loads occur, using weather telemetry (Temp, Humidity, Solar Irradiance).
- **Autonomous Transition Controller:**
  Seamlessly shifts between Renewable Storage Mode and Main Grid Mode, buffering peak loads with zero human intervention.
- **High-Availability Fail-Safe Architecture:**
  Dual-tier inference combining Zoho QuickML API with an onboard Random Forest fallback, guaranteeing 100% operational uptime.
- **Dynamic BESS Simulation (500 kWh):**
  Simulates realistic battery state-of-charge (SoC) with 10%-100% safety bounds, 30% peak absorption, and off-peak recharge.

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End-to-End System Architecture

1. TELEMETRY INGESTION

CSV Upload & Interactive Form

Extracts timestamps, cleans missing values with median imputation, and engineers hour & dayofweek features.

2. FORECASTING ENGINE

QuickML + High-Availability RF

Primary: Zoho QuickML REST API.

Fallback: Local Random Forest Regressor (RMSE 13.35 kW) ensuring continuous 24h predictions.

3. DISPATCH CONTROLLER

Autonomous Switching Logic

Evaluates hourly demand (>160 kW) and evening peak window (18:00-22:00) to trigger Renewable Storage Mode.

4. BESS SIMULATION

500 kWh Battery Storage

Tracks hourly State-of-Charge (SoC), discharging 30% of load during peak and charging at 40 kWh/h during off-peak.

5. CATALYST APPSAIL

Interactive Web Dashboard

Alpine.js + Chart.js dashboard deployed on Catalyst with Overview, Grid Modes, Battery SoC, and Logs.

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ML Demand Forecasting & Fail-Safe Architecture

FEATURE ENGINEERING & MODELING

Telemetry Feature Extraction Pipeline

hour (0-23)

Captures distinct diurnal load cycles (daytime activity vs night)

dayofweek (0-6)

Distinguishes industrial/commercial weekdays from weekend baselines

ambient_temp_c

Primary thermal cooling load driver (air conditioning surge)

humidity_pct

Latent heat factor influencing HVAC electrical draw

solar_irradiance_wm2

Direct proxy for behind-the-meter solar offset capacity

Model Architecture: Random Forest Ensemble

Trained on 4,000 historical rows with 300 estimators, min_samples_leaf=3.

Achieved Holdout RMSE: 13.35 kW (QuickML Benchmark: ~12.0 kW).

HIGH AVAILABILITY & QUOTA RESILIENCE

Zero-Downtime Prediction Architecture

- **Zoho QuickML Quota Exhaustion:**
  Free-tier limits (500 calls) trigger HTTP 403 PREDICTION_USAGE_LIMIT_EXCEEDED during bulk or repeated evaluations.
- **The 'Probe-First' Pattern:**
  predict.py sends 1 single probe request to check QuickML availability.
  If accessible, it processes remaining rows via ThreadPoolExecutor.
- **Instant Autonomous Fallback:**
  If QuickML returns 403, network failure, or timeout, the engine instantaneously switches to the local RF model without dropping a single row.
- **Self-Healing Retraining:**
  If pre-pickled model_local.joblib is ever missing, predict.py auto-retrains from train_prepared.csv in <2 seconds.
- **Pre-Submission Verification:**
  Evaluated against 1,000 test rows via test_eval.py with 100% SUCCESS.

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Transition Controller & Dispatch Logic

CONDITION A

Renewable Storage Mode

Trigger Criteria:

* Demand > 160 kW
* OR Hour in [18:00, 22:00] (Inclusive)

Microgrid Dispatch Action:

Switches power source from Main Grid to localized BESS battery storage. Discharges 30% of hourly load (kW) directly from the 500 kWh battery. Reduces peak utility power draw, lowering stress on substation lines.

CONDITION B

Main Grid Mode

Trigger Criteria:

* Demand <= 160 kW
* AND Hour outside 18:00-22:00

Microgrid Dispatch Action:

Microgrid draws baseline load directly from primary utility grid. Activates off-peak BESS recharging at steady rate of 40 kWh / hour. Replenishes SoC by +8.0% / hour capped at 100% capacity limit.

SAFETY OVERRIDE

Critical Overload Alert

Trigger Criteria:

* Predicted Demand >= 200 kW
* Triggered INDEPENDENTLY of Mode

Emergency Protocol:

Flashes high-priority UI alert banner. Emits instant log tag in telemetry. Pre-arms secondary load-shedding relays. Forces BESS maximum discharge rate. Protects transformer health and prevents cascading blackout shutdown.

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Battery Energy Storage System (BESS) Simulation

BESS MODEL SPECIFICATIONS

Total Battery Capacity:
500 kWh (High-Density Lithium Iron Phosphate)

Initial State of Charge:
80.0% (400 kWh reserve at simulation start)

Operational SoC Boundaries:
10.0% Minimum (Floor) to 100.0% Maximum (Ceiling)

SoC Safety Buffer:
Protects cells against deep discharge damage (<10%)

Mathematical Simulation Formulary

Discharge Rate (Renewable Storage Mode):

Delta SoC = - [(0.30 x Demand_kW) / 500 kWh] x 100%

Example: 180 kW demand discharges 54 kWh (-10.8% SoC/hr).

Recharge Rate (Main Grid Mode):

Delta SoC = + [40 kWh / 500 kWh] x 100% = + 8.0% / hr

Replenishes battery up to maximum 100.0% limit.

SIMULATED 24-HOUR BESS PROFILE

State of Charge (SoC) Dynamics

Representative 24-Hour Cycle from sample_test_input.csv:

Hour	Demand	Active Mode	SoC Change	Net SoC
00:00	125 kW	Main Grid	+8.0% (Charge)	88.0%
04:00	110 kW	Main Grid	+8.0% (Charge)	100.0% (Max)
12:00	175 kW	Renewable Storage	-10.5% (Discharge)	89.5%
15:00	190 kW	Renewable Storage	-11.4% (Discharge)	78.1%
19:00	215 kW	Renewable + Alert	-12.9% (Discharge)	52.3%
21:00	195 kW	Renewable Storage	-11.7% (Discharge)	28.9%
23:00	130 kW	Main Grid	+8.0% (Charge)	36.9%

*** Guaranteed Continuous Operation:**

Battery remains within safe 10-100% operational bounds throughout 24-hour cycle.

Zoho Catalyst AppSail Dashboard & UI

VIEW 1: SYSTEM OVERVIEW

Dual-Axis Forecasting Chart

* Real-time KPI Metric Cards (Active Load, Grid Status, Alert Counter).
* Dual-Axis Forecast Chart: Predicted Demand (kW) vs. BESS Battery SoC (%).
* Instant 24h, 48h, 7d Forecast Horizon Toggle.
* Glassmorphism styling with dark mode and zero UI latency.

VIEW 2: DISPATCH CONTROLLER

Smart Grid Switching Analytics

* Visual breakdown of Renewable Storage Mode vs. Main Grid Mode hours.
* Threshold indicator cards highlighting >160 kW and 18-22h evening peak.
* Transition frequency monitor and peak-shaving summary.
* Operational status chips with dynamic CSS pulsing badges.

VIEW 3: BESS BATTERY MONITOR

500 kWh State-of-Charge Dynamics

* High-precision battery gauge showing real-time SoC percentage.
* Hourly discharge and charge trajectory timeline.
* Safety threshold alerts for low-battery (<15%) and full reserve (100%).
* Simulated energy savings calculation vs. standard utility tariff.

VIEW 4: REAL-TIME TELEMETRY LOG

Filterable Table & CSV Export

* Complete tabular display of every forecast hour.
* Search by timestamp and filter by Grid Mode or Overload status.
* Ingest custom CSV telemetry or edit individual hour values.
* One-click CSV export of predictions and control signals.

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Impact, Evaluation Results & Future Scope

HACKATHON DELIVERABLES & RESULTS

- **Evaluation Scorecard:**
  Passed test_eval.py with 100% SUCCESS across 1,000 hidden test rows.
- **Forecasting Accuracy:**
  Holdout RMSE of 13.35 kW on 800-row holdout set, achieving competitive parity with QuickML AutoML (~12.0 kW).
- **Cloud Deployment:**
  Fully hosted and live on Zoho Catalyst AppSail with zero configuration required by end users.
- **Security Compliance:**
  Zero credentials in source control; .gitignore safeguards secrets and local model binaries.
- **Peak Load Reduction:**
  30% peak demand shaving successfully reduces distribution transformer strain during 18:00-22:00 peak hours.

Repository: https://github.com/Koushik554/Bits-Hackthon

Live Web App: Zoho Catalyst AppSail (Microgrid Service)

FUTURE HORIZONS & ROADMAP

- **Multi-Battery BESS Swarm Coordination:**
  Scale from a single 500 kWh pack to a distributed network of commercial and residential batteries across the microgrid feeder.
- **Real-Time IoT Smart Meter Integration:**
  Direct MQTT telemetry streaming via Zoho Catalyst IoT connectors for sub-minute dispatch adjustments.
- **Dynamic Real-Time Electricity Tariffs:**
  Incorporate spot market power pricing to maximize battery charge during negative/cheap tariff windows and discharge during high-rate surges.
- **Automated Demand Response:**
  Issue automated load-shedding commands to smart industrial appliances during Critical Overload Alerts (>=200 kW).

Thank You, HackSphere 3.0 Organizers & BITS Pilani!

Ready for Q&A and Live Demo Evaluation.

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